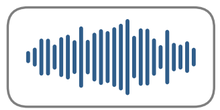
current_time **2026-11-08 07:10** user **Chị Hoa**

14 installed apps and their tool schemas

history past tool calls and results

val **70.91** public **58.02**

3.2 s per sample, limit 30 s



Request audio

6 s, 16 kHz



history audio

Gipformer

RNN-T, 68M

WER on val audio

base 17.1%

ours **10.7%**

sherpa-onnx, CPU

ASR

***fine-tuned** on TRAIN audio*

*"**trả tiền điện** bằng **ví điện tử** giúp mình ví mà **không được** thì trừ vào **tài khoản ngân hàng** nhé"*

gold: **không đủ**

Transcript

***lower-case**, no punctuation*

rules

tool schemas

context

history

request

Prompt

*≈ **6k** tokens, ≤ 16k*

Qwen3.5-4B

+ LoRA r16

vLLM

LLM

***SFT** on gold and ASR text*

tool result → *next LLM turn*

schema

x-ref IDs seen

no repeats

rejected:

error to the LLM

Validate

***before** env.call*

ToolEnv *zaic_tools*

1 `bill_payment.list_bills()`

→ điện **845.000đ**, bill_02

2 `ewallet.get_balance()`

→ **713.606đ** < 845.000đ

3 `bill_payment.pay(bill_02, source="bank")`

***stop** when the LLM calls nothing*

Agent loop

***≤ 10** turns, one env per sample*

3 calls

in order

`state_hash`

missing slot

ask back

Output